Delay Discounting Experiment

PSY310 Lab in Psychology

Lab Report

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GitHub Link :

Introduction

Intentional binding is the phenomenon where individuals perceive the temporal interval between their voluntary action and the resulting sensory consequence as shorter than it actually is—this "temporal compression" occurs primarily in contexts where the action is believed to be voluntary and causally linked to its outcome.

Intentional binding serves as an implicit measure of the sense of agency: the subjective experience that one's actions cause external events. For example (here), when a person presses a button and a tone sounds after a short delay, they often report the events as occurring closer together in time when they feel responsible for causing the tone. Participants then estimate the interval between their action and the effect. Intentional binding is quantified by comparing their estimates in voluntary versus passive or baseline conditions.

This paradigm enables the study of sense of agency and underlying cognitive processes, with PsychoPy providing tools to easily implement such designs and record interval estimates for subsequent analysis.

Method

Participants

The experiment was performed by one participant - 19 years

old and a female undergraduate student studying at Ahmedabad University. 400 trials were conducted.

Materials & Procedure

The experiment was created on a laptop, using the PsychoPy v2025.1.1. 100 trials were conducted per participant.

The experiment had 3 routines -a trial routine, a beep routine and a response routine. For the first routine, we had three components and two polygons (Figure 1 and 2) and a keyboard response (space bar). The circles were displayed one after the other; the first circle was white in color and the second one was green.

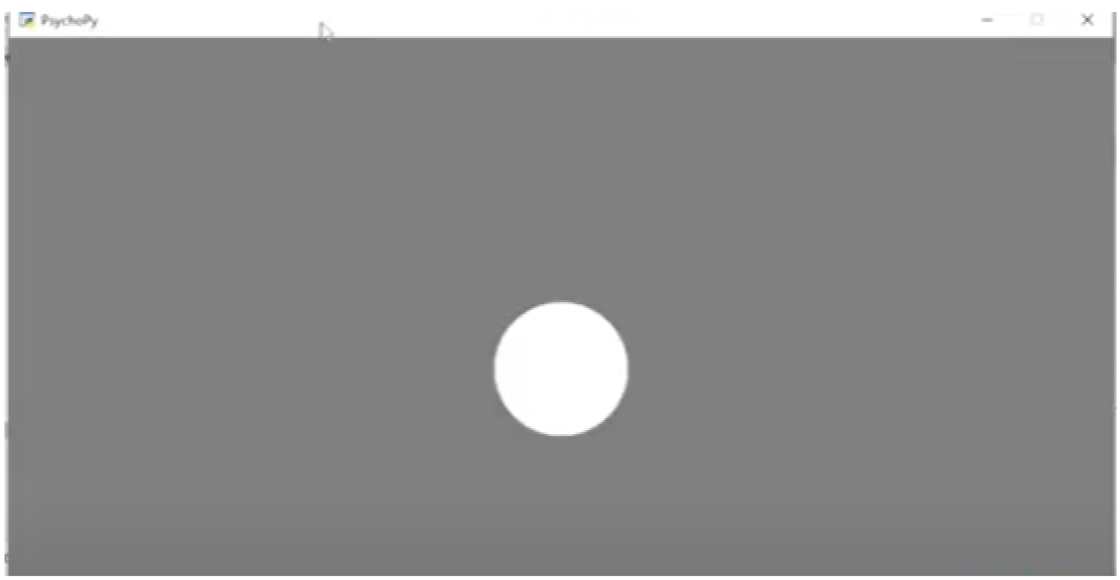

Figure 1

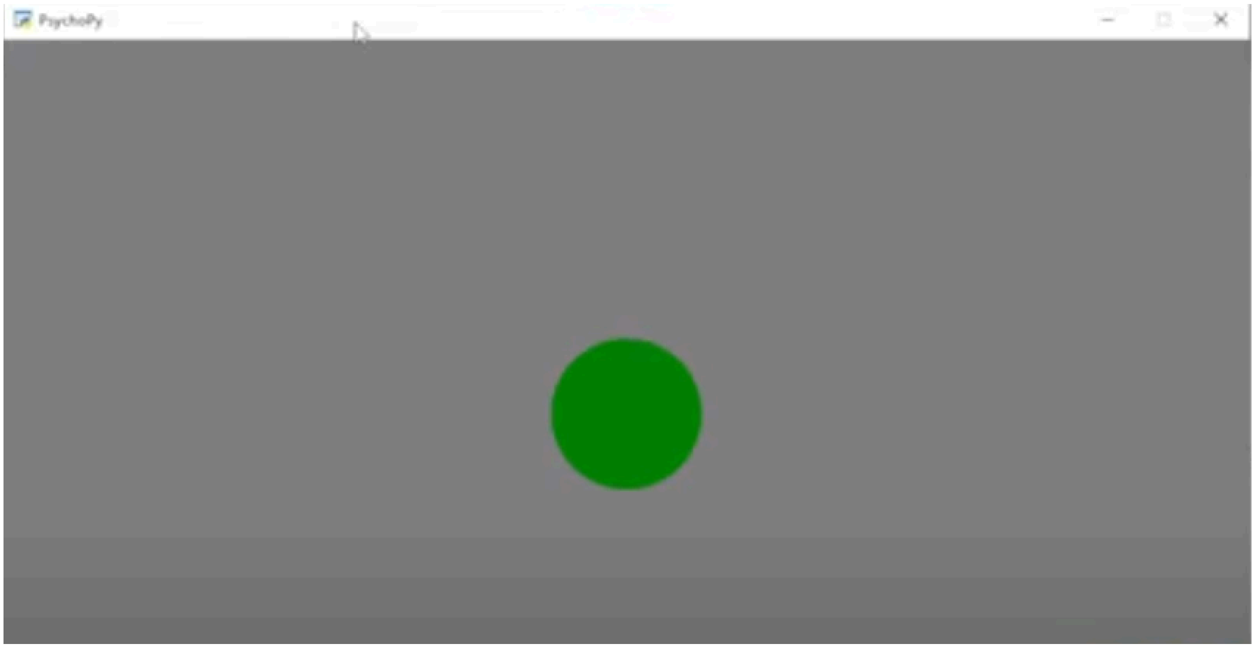

Figure 2

After the green circle appears, we click the space bar and we hear a beep sound. The sound (beep routine) was added with a 0.3 duration and starting value as “$delay”. Followed by the last routine, the response routine had three components - a text, a textbox and a button, this is where we would record our responses (Figure 3). A loop was added upon the three routines and set at random.

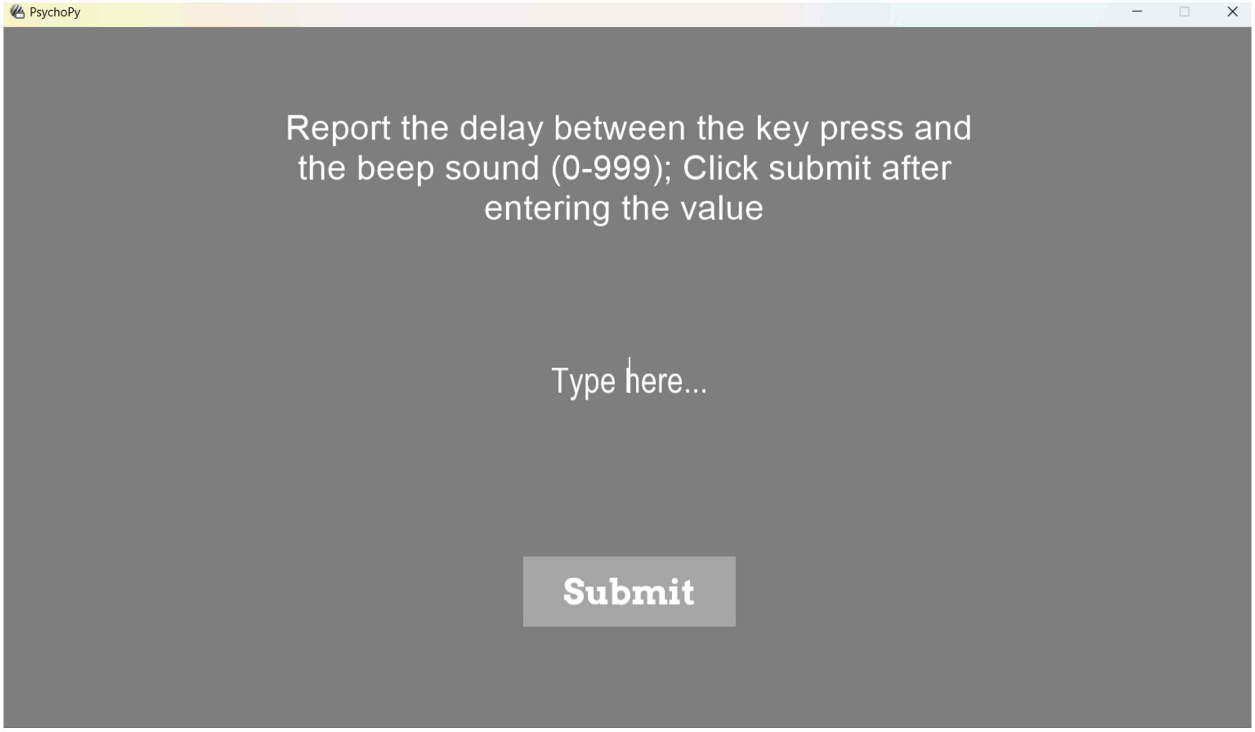

Figure 3

After this, we run the experiment and the data will be automatically saved in an excel file.

Results

In the expected condition, the average estimated interval was 281.62 ms, whereas it was 269.58 ms in the unexpected condition.

Discussions

The benefit of implicit measures of sense of agency is that they eliminate the demand characteristics and also social desirability because it does not involve any self-reports but rather reflects automatic, behavioural reactions including reaction-time variabilities. This enables the researcher to access fine processes of prediction that the subjects might not be aware of. The small RT contrast between the predicted ([?]281 ms) and the unexpected ([?]270 ms) intervals in your data indicate that temporal prediction effects are not large but rather small - an example of how the implicit measures can be used to demonstrate underlying processes even when they are weak. Yet these measures are hard to interpret due to the fact that the reaction times are indicative of a range of other cognitive variables other than the agency and are generally quite varied and demand a high level of accuracy in timing. Accordingly, although the implicit tasks can offer significant behavioural information, its result must be viewed with caution and preferably complemented with direct actions.

References

1. Gutzeit, J., Weller, L., Kürten, J., & Huestegge, L. (2023). Intentional binding: Merely a procedural confound?. *Journal of Experimental Psychology: Human Perception and Performance*, *49*(6), 759.
2. Obhi, S. S., & Hall, P. (2011). Sense of agency and intentional binding in joint action. *Experimental brain research*, *211*(3), 655-662.